PODC '24 · PAPER PRESENTATION

DeepSpeed Ulysses

System optimizations for training extreme long-sequence Transformer models

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Why long sequences matter

- **Generative AI**

Conversational AI needs **long chat histories** to stay coherent

Book- and chapter-level summarization spans tens to hundreds of thousands of words

Multimodal models reason over video, speech and waveforms — high-dimensional, long inputs

AI for Science

Genome language models: the human genome is **6.4 billion letters**

Climate & weather models need sub-kilometer, high-resolution inputs

Healthcare models conditioned on an **entire patient record**

THE GAP

Four dimensions of LLM computation — one is unsolved

01 · BATCH SIZE

Data parallelism

Replicate the model, split the batch. Well studied.

02 · HIDDEN DIM

Tensor parallelism

Split operators within a layer across devices.

03 · LAYERS

Pipeline parallelism

Split the model depth-wise into stages.

04 · SEQUENCE LENGTH

Sequence parallelism

The dimension long-context training actually needs — and the one prior systems handle poorly.

PROBLEM STATEMENT

Where existing parallelism falls short

- Data, tensor and pipeline parallelism **cannot scale along the sequence dimension** at all
- Existing sequence-parallel methods are **communication-inefficient** — cost grows with sequence length
- They often demand **intrusive, error-prone code rewrites** to adopt

METHOD	COMM.	PARAM. MEM.	ANY ATTENTION
ColAI-SP	O(M)	✗	✗
Megatron-SP	O(M)	✓	✓
Ulysses	O(M/P)	✓	✓

M = message size · P = number of GPUs. Lower communication complexity is better.

THE IDEA

Partition the sequence across GPUs — then use all-to-all to switch into head-parallel attention, and back again.

Each GPU holds N/P tokens. One all-to-all gives every GPU the full sequence but only a subset of attention heads; a second all-to-all returns to sequence parallelism for the rest of the layer.

How Ulysses works

SEQUENCE PARALLEL

N/P tokens per GPU

All heads, a slice of the sequence. Project to Q, K, V.

G0 G1 G2 G3

ALL-TO-ALL →

HEAD PARALLEL

Full sequence, fewer heads

Local full attention per head — any kernel: dense, sparse, FlashAttention.

h0 h1 h2 h3

ALL-TO-ALL →

SEQUENCE PARALLEL

Back to N/P per GPU

Re-partition for MLP, layer norm and the rest of the block.

G0 G1 G2 G3

CONTRIBUTION · WHY IT SCALES

Communication stays constant as you scale

ULYSSES — ALL-TO-ALL

4Nd / P

Each link carries 1/P of the data. Scale N and P together → volume per GPU is **constant**.

MEGATRON / COLAI — ALL-GATHER

4Nd

Independent of P — **P× larger** , and keeps growing as sequences get longer.

Result: **over 10× communication reduction** versus the existing state of the art.

EVIDENCE · MICROBENCHMARK

All-to-all vs. all-gather + reduce-scatter

METHOD	COLLECTIVE	DATA MOVED	TOTAL COMM. TIME
Megatron-LM	all-gather + reduce-scatter	8 GB	34 ms
Ulysses	all-to-all	8 GB	4.9 ms

~7× faster for the same volume. Each GPU pair only exchanges its 1/P slice — no contention.

256K sequence · 4K hidden · 8× A100

Three things that make it practical

MEMORY

ZeRO-3 integration

Model states partitioned across the combined data + sequence parallel groups — scale model size *and* sequence length together.

FLEXIBILITY

Attention-agnostic

Dense, sparse, causal and FlashAttention v2 all drop in — attention stays an ordinary local full-attention op.

ADOPTION

Easy & portable

Minimal code changes. Open-sourced in DeepSpeed and Megatron-DeepSpeed, already in production use.

Experimental setup

Cluster	ThetaGPU · Argonne NL
GPUs / node	8 × A100 · 40 GB
Inter-node	HDR200 IB · 200 GB/s
GPT-7B	d 4096 · 32 heads · 32 layers

Node	NVIDIA DGX A100
Intra-node	NVLink / NVSwitch
Architecture	Standard GPT
GPT-30B	d 6144 · 64 heads · 64 layers

RESULT · SEQUENCE SCALABILITY

1M+

token context length on a 1.2B GPT model

Sequence length scales **linearly with GPU count** while sustaining roughly constant throughput across lengths — context is no longer bounded by a single GPU's memory.

RESULT · VS. STATE OF THE ART

Faster, longer, leaner than prior methods

2.5×
higher training throughput than the SOTA baseline

4×
longer sequences than existing systems

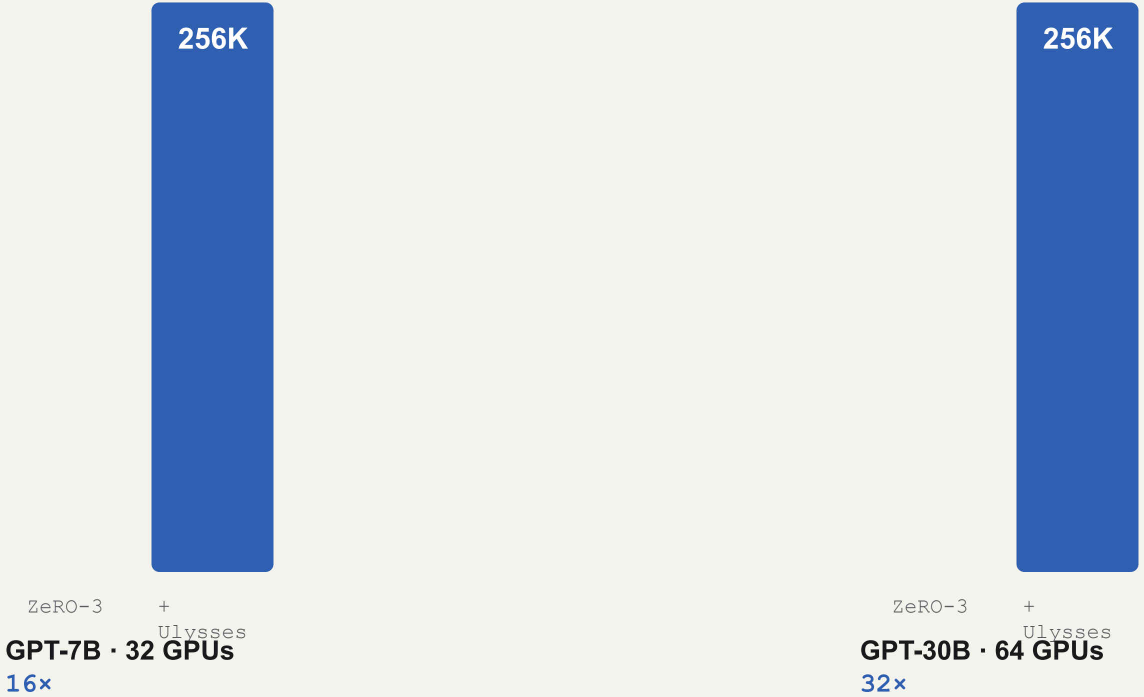
10×
less communication volume

175+
TFLOPs/GPU sustained — over 54% of A100 peak

Compared against Megatron-LM and ColAI-SP sequence parallelism on GPT-7B (32 GPUs) and GPT-30B (64 GPUs), dense and sparse attention.

RESULT · MEMORY

Max sequence length: ZeRO-3 alone vs. + Ulysses



ZeRO-3 alone tops out at 16K (7B) and 8K (30B) before activations overflow. ZeRO partitions weights; Ulysses partitions activations.

Training ClimaX at higher resolution

ClimaX is a Vision-Transformer climate model. Higher-resolution images mean **quadratically longer** token sequences — and quick OOM.

- **Baseline:** 768×768 image → 108K tokens, then out of memory
- **With Ulysses:** 1536×1536 image → 432K tokens
- Same 64 A100 GPUs, lower peak memory

4×

longer sequences on the very same hardware

SUMMARY

Contributions & takeaways

- A novel **all-to-all sequence parallelism** with communication that stays constant as you scale
- **ZeRO-3 integration** for memory, plus **attention-agnostic** design and easy adoption
- First to reach **million-token** context with sustained, near-peak efficiency

THE HEADLINE

2.5× faster training, 4× longer sequences

than the existing state-of-the-art baseline.

Questions to open up

1. Constant communication holds only when **N and P scale together** . On a fixed GPU budget with ever-growing sequences, does the advantage still hold?

2. Parallelism degree is **capped by the number of attention heads** . How limiting is that for models with few heads, or with grouped-query / multi-query attention?

3. How does Ulysses relate to **ring-attention and context-parallel** methods that appeared since — and could they be combined?